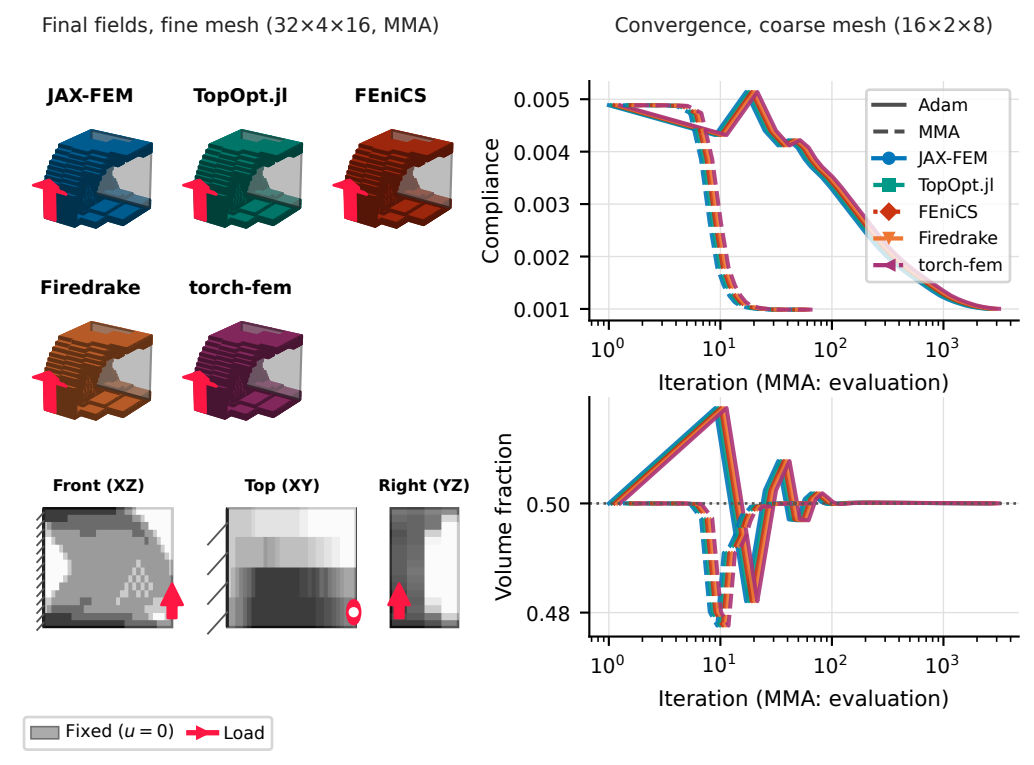

Final fields, fine mesh (32×4×16, MMA)
JAX-FEM
TopOpt.jl
FEniCS
Firedrake
torch-fem
Front (XZ)
Top (XY)
Right (YZ)
Fixed ($u = 0$)
Load
Convergence, coarse mesh (16×2×8)
Compliance
0.005
0.004
0.003
0.002
0.001
10^0
10^1
10^2
10^3
Iteration (MMA: evaluation)
Adam
MMA
JAX-FEM
TopOpt.jl
FEniCS
Firedrake
torch-fem
Volume fraction
0.50
0.48
10^0
10^1
10^2
10^3
Iteration (MMA: evaluation)